

AI or Not?

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

My starting explanation

2. Practise with fading help

Use W2: Complete the photo library explanation for Evidence and decision mechanisms. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Evidence and decision mechanisms. Proposed explanation: “All rules are outside AI”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Investigate a new hybrid system

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A written rule displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent. Identify each mechanism and explain why a correct CHECK PLANTS message would not prove an accurate temperature prediction.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A bus displays a delay prediction learned from past journeys. A written rule turns the display red above 10 minutes. A driver can override the message. Identify all three mechanisms. Would seeing a red display

prove that learning occurred?

6. Solve the given case

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Classify all six using mechanism evidence. Which two need more information?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a learned model? What evidence supports each classification?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 1 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or learned model; justify the smallest safe solution.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A bus displays a delay prediction learned from past journeys. A written rule turns the display red above 10 minutes. A driver can override the message. Identify all three mechanisms. Would seeing a red display prove that learning occurred?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Separate observations from explanations

A door opens when a visitor approaches. Its label says intelligent entrance. List two observations and write one mechanism question whose answer would help classify the opening decision.

My reasoning and evidence

13. W2 Complete the photo library explanation

Use the photo-library case from the lesson. Classify the prediction, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

My reasoning and evidence

14. W3 Challenge an unsupported conclusion

Maya says a vacuum must learn because it changes direction after hitting a chair. Give two different mechanisms consistent with that observation. What evidence would distinguish them?

My reasoning and evidence

15. W4 Investigate a new hybrid system

A school greenhouse estimates tomorrow’s temperature using a model fitted on historical weather. A written rule displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent. Identify each mechanism and explain why a correct CHECK PLANTS message would not prove an accurate temperature prediction.

My reasoning and evidence

16. W5 Design an evidence card

Invent one product containing two mechanisms. State the supplied facts, draw an information path and write one claim the facts support and one they do not. Do not use real classmates' personal information.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: AI or Not?

A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

A weather model was fitted on earlier weather records; its forecast is 70% rain.

A written rule shows TAKE ONE at rain forecasts of 60% or more. The teacher decides whether to distribute umbrellas.

AI ACADEMY / AI-1/1

AI or Not?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Weather records

Earlier observations were used to fit the forecasting model.



2. Forecast

The learned model outputs 70% rain, not a guaranteed shower.



3. Display rule

70 is at least 60, so the written rule displays TAKE ONE.



4. Final decision

The teacher checks current conditions before distributing umbrellas.

CHANGE ONE THING

Keep the forecast but raise the display boundary to 80%.

Explain the changed result using the matching lesson.

Point to the learned stage. Does the changed display prove the weather forecast changed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The teacher checks current conditions before distributing umbrellas.

1. Weather records
2. Final decision
3. Display rule
4. Forecast

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Final decision The teacher checks current conditions before distributing umbrellas. The fitting record supports calling the forecasting stage learned. Comparing 70 with 60 is an explicit rule. The final action belongs to a person. Merely seeing TAKE ONE would not reveal any of these mechanisms; the process notes supply that evidence.

Practice 2: explain the deciding evidence

Changed-case practice: Keep the forecast but raise the display boundary to 80%. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The display no longer says TAKE ONE.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The display no longer says TAKE ONE. 70 is below 80; changing a display rule does not refit the model.

Repair a missing building block

This is the starting lesson. Ask your teacher to model one observation, one written rule and one deciding step, then try your own example.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
